

# AP Calculus BC

## AP Մաթեմատիկական անալիզ BC

|              |                                                                                    |
|--------------|------------------------------------------------------------------------------------|
| COURSE CODE  | MTH-APCB                                                                           |
| PROGRAM      | Advanced Placement and Examination Preparation · Առաջադեմ և քննական պատրաստություն |
| LEVEL        | Typically Grade 11 or 12 · College Board aligned                                   |
| DURATION     | 144–162 instructional hours (approx. 115 lessons) plus examination preparation     |
| PREREQUISITE | Precalculus (MTH-PC) with strong performance, or AP Calculus AB (MTH-APCA).        |
| LEADS TO     | University Calculus III, Linear Algebra, or Differential Equations                 |

### COURSE DESCRIPTION

Դասընթացի նկարագրություն

AP Calculus BC covers the equivalent of two semesters of university calculus. It contains all eight AB units and adds two more: parametric equations, polar coordinates, and vector-valued functions; and infinite sequences and series. It also extends several AB topics — additional integration techniques, improper integrals, logistic differential equations, Euler's method, and arc length. Students receive an AB subscore in addition to the BC score.

### TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

BC is not AB with extra chapters; it is a faster course that assumes the student converts new technique into fluency quickly. Վարժարան therefore front-loads the AB material at an accelerated pace to protect adequate time for Units 9 and 10, which together account for roughly a third of the examination and are where most BC students lose points. Series in particular rewards a systematic approach: students are taught a decision procedure for selecting a convergence test and are required to name the test and verify its conditions before applying it, rather than trying tests until one seems to work.

### PLACEMENT AND READINESS INDICATORS

Ընդունելություն և պատրաստվածություն

The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

| READINESS INDICATOR                                                                     | WHAT IT TELLS THE TEACHER                                                                           |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Completed AB material, or demonstrates strong Precalculus fluency and rapid acquisition | BC pacing is unforgiving; a student who needs repetition to acquire technique should take AB first. |

| READINESS INDICATOR                                                         | WHAT IT TELLS THE TEACHER                                                                            |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Manipulates rational and trigonometric expressions with automaticity</b> | Required for the additional integration techniques in Unit 6 and for series manipulation in Unit 10. |
| <b>Works comfortably with parametric and polar representations</b>          | Covered in Precalculus Unit 6; if absent, add a bridge before Unit 9.                                |
| <b>Writes complete mathematical justifications</b>                          | As in AB, the largest single source of avoidable score loss.                                         |
| <b>Works with sequences, series, and sigma notation</b>                     | Precalculus Unit 7; a strong predictor of Unit 10 performance.                                       |

## COURSE LEARNING OUTCOMES

Ուսումնական արդյունքներ

On completion of this course the student will be able to:

- 1 All AP Calculus AB outcomes, at BC pace and depth.
- 2 Integrate using integration by parts and partial fractions.
- 3 Evaluate improper integrals and determine their convergence.
- 4 Solve logistic differential equations and interpret their behaviour.
- 5 Apply Euler's method to approximate solutions to differential equations.
- 6 Compute arc length and the area of a surface of revolution.
- 7 Differentiate and integrate parametric and vector-valued functions.
- 8 Analyze motion in the plane using vector-valued functions.
- 9 Compute derivatives and areas in polar coordinates.
- 10 Determine convergence or divergence of sequences and series using appropriate tests.
- 11 Apply the alternating series error bound and the Lagrange error bound.
- 12 Determine the radius and interval of convergence of a power series.
- 13 Represent functions as Taylor and Maclaurin series and manipulate known series.

## COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

### UNIT 1 • 10-11 LESSONS • APPROX. 4-7% OF THE EXAMINATION

## Limits and Continuity

Սահմաններ և անընդհատություն

### TOPICS COVERED

- Limit notation, estimation from graphs and tables, and algebraic determination
- Selecting procedures for determining limits
- The squeeze theorem
- Types of discontinuity and continuity at a point and over an interval
- Removing discontinuities
- Infinite limits and vertical asymptotes
- Limits at infinity and horizontal asymptotes
- The Intermediate Value Theorem

### LEARNING OBJECTIVES

- 1 Determine limits by an appropriately selected and justified technique.
- 2 Justify continuity using the three-part definition.
- 3 Apply the Intermediate Value Theorem with hypothesis verification.

### MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Determines limits at 92% accuracy.
- Writes complete continuity justifications in 9 of 10 trials.
- Verifies theorem hypotheses in writing in 9 of 10 trials.

### COMMON MISCONCEPTIONS AND TEACHER RESPONSE

| MISCONCEPTION                                                          | RECOMMENDED RESPONSE                                                                           |
|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Limit notation is dropped before the limit is evaluated.</b>        | It must remain until evaluation. This is a scored notation point on the free-response section. |
| <b>A theorem conclusion is stated without hypothesis verification.</b> | Verification is worth points independently of the conclusion. Establish the habit in Unit 1.   |

#### TEACHING NOTE

*BC students typically find this unit straightforward. Use the pace advantage to establish the written-justification standard rather than to move ahead — the standard, once set, applies to the whole year.*

## UNIT 2 • 10-11 LESSONS • APPROX. 4-7% OF THE EXAMINATION

**Differentiation: Definition and Fundamental Properties**

Ածանցում. սահմանում և հիմնական հատկություններ

**TOPICS COVERED**

- Average and instantaneous rates of change; the derivative definition and notation
- Estimating derivatives; differentiability and continuity
- Power, constant, sum, difference, and constant multiple rules
- Derivatives of trigonometric, exponential, and logarithmic functions
- Product and quotient rules
- Interpreting the derivative in context with units

**LEARNING OBJECTIVES**

- 1 Compute derivatives from the definition and recognize limit expressions as derivatives.
- 2 Differentiate combinations of elementary functions.
- 3 Interpret a derivative in context with units and a complete sentence.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Differentiates at 94% accuracy using fundamental rules.
- Recognizes derivative-form limits in 9 of 10 trials.
- Writes contextual interpretations with units in 9 of 10 trials.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                    | RECOMMENDED RESPONSE                                                                                                                  |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Speed of acquisition is mistaken for fluency.</b>             | BC students often understand a rule immediately and still execute it slowly. Fluency probes measure the difference; do not skip them. |
| <b>Applied interpretation is treated as trivial and skipped.</b> | It is scored on every free-response set. Require it from the first assignment.                                                        |

**TEACHING NOTE**

*The chief risk in the early BC units is complacency. Maintain fluency probes even when conceptual understanding is obviously secure — Unit 10 will demand automaticity in everything preceding it.*

## UNIT 3 • 9-10 LESSONS • APPROX. 4-7% OF THE EXAMINATION

**Differentiation: Composite, Implicit, and Inverse Functions**

Ածանցում. բարդ, անուղղակի և հակադարձ ֆունկցիաներ

**TOPICS COVERED**

- The chain rule
- Implicit differentiation
- Differentiating inverse functions
- Differentiating inverse trigonometric functions
- Selecting procedures for calculating derivatives
- Higher-order derivatives

**LEARNING OBJECTIVES**

- 1 Apply the chain rule to nested compositions of any depth.
- 2 Perform implicit differentiation and find tangent lines to implicit curves.
- 3 Differentiate inverse and inverse trigonometric functions.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Differentiates at 92% accuracy including multi-level chain rule.
- Performs implicit differentiation correctly in 9 of 10 trials.
- Selects efficient procedures in 9 of 10 trials.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                                  | RECOMMENDED RESPONSE                                                                               |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>The chain rule is omitted at the innermost level of a deep composition.</b> | Write the composition structure before differentiating. Depth is where BC problems differ from AB. |
| <b>Inverse trigonometric derivative forms are confused with each other.</b>    | Derive them by implicit differentiation once rather than memorizing six forms.                     |

**TEACHING NOTE**

*Deriving the inverse trigonometric derivatives by implicit differentiation, rather than presenting a table, halves the memorization and gives the student a recovery route when memory fails.*

## UNIT 4 • 10-11 LESSONS • APPROX. 6-9% OF THE EXAMINATION

**Contextual Applications of Differentiation**

Ածանցյալի կիրառական խնդիրներ

**TOPICS COVERED**

- Interpreting the derivative in context
- Straight-line motion: position, velocity, acceleration, speed
- Rates of change in applied contexts
- Related rates problems
- Local linearity and linearization
- L'Hôpital's rule for indeterminate forms

**LEARNING OBJECTIVES**

- 1 Solve related rates problems from described situations.
- 2 Analyze straight-line motion and justify when speed increases.
- 3 Apply L'Hôpital's rule after verifying the indeterminate form.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Solves related rates problems in 4 of 5 tasks.
- Analyzes motion including speed reasoning in 9 of 10 trials.
- Verifies indeterminate forms before applying L'Hôpital's rule in 9 of 10 trials.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                                                  | RECOMMENDED RESPONSE                                               |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>L'Hôpital's rule is applied repeatedly without re-checking the form.</b>                    | Re-verify after each application. The form may become determinate. |
| <b>Speed reasoning is attempted without separating the signs of velocity and acceleration.</b> | State both signs explicitly before concluding.                     |

**TEACHING NOTE**

*BC problems in this unit often combine related rates with parametric motion later in the course. Flag the connection when Unit 9 arrives.*

## UNIT 5 • 12-13 LESSONS • APPROX. 8-11% OF THE EXAMINATION

**Analytical Applications of Differentiation**

Ածանցյալի վերլուծական կիրառություններ

**TOPICS COVERED**

- The Mean Value Theorem and the Extreme Value Theorem
- Critical points, increasing and decreasing intervals
- The first derivative test and the candidates test
- Concavity and the second derivative test
- Sketching functions and their derivatives
- Connecting  $f$ ,  $f'$ , and  $f''$
- Optimization problems
- Behaviours of implicit relations

**LEARNING OBJECTIVES**

- 1 Justify extrema and concavity in AP rubric language.
- 2 Connect the graphs of  $f$ ,  $f'$ , and  $f''$  with justification.
- 3 Solve optimization problems with verification and contextual answers.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Produces complete written justifications in 9 of 10 trials.
- Connects  $f$ ,  $f'$ , and  $f''$  correctly in 4 of 5 tasks.
- Solves optimization problems with verification in 4 of 5 tasks.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                                         | RECOMMENDED RESPONSE                                                   |
|---------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Justification is abbreviated because the conclusion is obvious to the student.</b> | The reader scores what is written. Obviousness is not a justification. |
| <b>Endpoints are omitted from the candidates test.</b>                                | Make the candidate list explicit and complete before evaluating.       |

**TEACHING NOTE**

*Justification quality established here carries directly into Units 9 and 10, where the reasoning is less familiar and the writing consequently harder. Do not let the standard slip.*

## UNIT 6 • 18-19 LESSONS • APPROX. 17-20% OF THE EXAMINATION

**Integration and Accumulation of Change**

Ինտեգրում և կուտակային փոփոխություն

**TOPICS COVERED**

- Accumulations of change and Riemann sums
- Summation and definite integral notation
- The Fundamental Theorem and accumulation functions
- Properties of definite integrals
- Antiderivatives and indefinite integrals
- Integrating using substitution
- Integrating functions using long division and completing the square
- Integration using integration by parts
- Integration using linear partial fractions
- Evaluating improper integrals
- Selecting techniques for antidifferentiation

**LEARNING OBJECTIVES**

- 1 Select and justify an integration technique before computing.
- 2 Integrate by parts, including with repeated application and the tabular method.
- 3 Decompose into linear partial fractions and integrate.
- 4 Evaluate an improper integral by writing it explicitly as a limit.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Selects the correct technique in 9 of 10 integrals.
- Integrates by parts and partial fractions at 90% accuracy.
- Evaluates improper integrals with explicit limits in 9 of 10 trials.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                                                              | RECOMMENDED RESPONSE                                                                                                       |
|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>An improper integral is evaluated without a limit.</b>                                                  | It is defined as a limit. Write the limit notation every time; its absence is a scored error.                              |
| <b>Integration by parts assignments of <math>u</math> and <math>dv</math> are made without a strategy.</b> | Use a systematic preference order for $u$ and check whether the resulting integral is simpler. If not, reverse the choice. |
| <b>Partial fractions is attempted on an improper rational function.</b>                                    | Divide first if the numerator degree is at least the denominator degree.                                                   |

**TEACHING NOTE**

*This is the highest-weighted unit on the BC examination and it carries the additional integration techniques. Allocate time proportionally — roughly one fifth of the course.*



## UNIT 7 · 10-11 LESSONS · APPROX. 6-9% OF THE EXAMINATION

**Differential Equations**

Դիֆերենցիալ հավասարումներ

**TOPICS COVERED**

- Modelling with differential equations and verifying solutions
- Sketching and reasoning with slope fields
- Approximating solutions using Euler's method
- Finding general and particular solutions by separation of variables
- Exponential models with differential equations
- Logistic models with differential equations

**LEARNING OBJECTIVES**

- 1 Apply Euler's method with a stated step size and assess the direction of the error.
- 2 Solve separable differential equations with initial conditions.
- 3 Analyze a logistic model: carrying capacity, the point of fastest growth, and long-run behaviour.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Applies Euler's method correctly in 9 of 10 trials.
- Solves separable equations at 92% accuracy.
- Analyzes logistic behaviour correctly in 4 of 5 tasks.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                                                 | RECOMMENDED RESPONSE                                                                                                    |
|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Euler's method is computed with the wrong step direction or an inconsistent step size.</b> | Tabulate the computation: $x$ , $y$ , $dy/dx$ , $\Delta y$ . The table structure prevents nearly all errors.            |
| <b>The logistic carrying capacity is confused with the inflection point value.</b>            | The inflection occurs at half the carrying capacity — the point of fastest growth. Derive it rather than memorizing it. |

**TEACHING NOTE**

*Logistic differential equations appear regularly on BC free-response sections and the qualitative questions — long-run behaviour, fastest growth — are often worth more than the algebra. Practice the reasoning, not only the solution.*

## UNIT 8 · 12-13 LESSONS · APPROX. 6-9% OF THE EXAMINATION

**Applications of Integration**

Ինտեգրալի կիրառություններ

**TOPICS COVERED**

- Average value of a function on an interval
- Connecting position, velocity, and acceleration with integrals
- Accumulation functions in applied contexts
- Area between curves in  $x$  and in  $y$ , including multiple intersections
- Volumes with cross-sections: squares, rectangles, triangles, semicircles
- Volume with the disc and washer methods
- The arc length of a smooth planar curve
- Distance travelled along a curve

**LEARNING OBJECTIVES**

- 1 Set up area and volume integrals with labelled diagrams.
- 2 Distinguish total distance from displacement.
- 3 Compute arc length for a smooth planar curve.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Sets up area and volume integrals correctly in 4 of 5 tasks.
- Computes arc length correctly in 9 of 10 trials.
- Distinguishes distance from displacement in 9 of 10 trials.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                                                        | RECOMMENDED RESPONSE                                                                                      |
|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>The arc length integrand is written without the square root or with the derivative unsquared.</b> | Derive the formula from the Pythagorean theorem on a differential element. Derivation prevents the error. |
| <b>Total distance is computed without splitting at the zeros of velocity.</b>                        | Locate the zeros first and split the integral. Otherwise the signed contributions cancel.                 |

**TEACHING NOTE**

*Arc length connects directly to parametric arc length in Unit 9. Present the two forms together when Unit 9 arrives so the student sees one idea, not two formulas.*

## UNIT 9 • 13-14 LESSONS • APPROX. 11-12% OF THE EXAMINATION

**Parametric Equations, Polar Coordinates, and Vector-Valued Functions**

Պարամետրական հավասարումներ, բևեռային կոորդինատներ և վեկտորական ֆունկցիաներ

**TOPICS COVERED**

- Defining and differentiating parametric equations
- Second derivatives of parametric equations
- Finding arc lengths of curves given parametrically
- Defining and differentiating vector-valued functions
- Integrating vector-valued functions
- Solving motion problems using parametric and vector-valued functions
- Position, velocity, acceleration, speed, and total distance in the plane
- Defining polar coordinates and differentiating in polar form
- Finding the area of a polar region
- Finding the area of the region bounded by two polar curves

**LEARNING OBJECTIVES**

- 1 Differentiate parametric equations, including the second derivative, correctly.
- 2 Analyze planar motion using vector-valued functions: velocity, speed, acceleration, and total distance.
- 3 Compute the area of a polar region and of a region between two polar curves.
- 4 Compute arc length for a parametrically defined curve.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Differentiates parametric equations at 90% accuracy including second derivatives.
- Solves planar motion problems in 4 of 5 tasks.
- Computes polar areas correctly in 9 of 10 trials with correct limits.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                                                                                        | RECOMMENDED RESPONSE                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>The second derivative of a parametric curve is computed as the second derivative of y with respect to t divided by that of x.</b> | It is the derivative of $dy/dx$ with respect to $t$ , divided by $dx/dt$ . Write the intermediate expression explicitly every time. |
| <b>Polar area limits are taken as 0 to <math>2\pi</math> by default.</b>                                                             | Determine the actual $\theta$ interval traced by the curve. Sketching first prevents this error, which is common and expensive.     |
| <b>Speed in the plane is computed as the derivative of position rather than as the magnitude of velocity.</b>                        | Speed is the magnitude of the velocity vector — a square root of a sum of squares.                                                  |

**TEACHING NOTE**

*Polar area limits are the most common error in this unit. Require a sketch showing the traced interval before any integral is written; the sketch is faster than the correction.*

## UNIT 10 • 18-19 LESSONS • APPROX. 17-18% OF THE EXAMINATION

**Infinite Sequences and Series**

Անվերջ հաջորդականություններ և շարքեր

**TOPICS COVERED**

- Defining convergent and divergent infinite series
- Working with geometric series
- The  $n$ th term test for divergence
- Integral test for convergence
- Harmonic series and  $p$ -series
- Comparison tests for convergence
- Alternating series test
- Ratio test for convergence
- Determining absolute or conditional convergence
- Alternating series error bound
- Finding Taylor polynomial approximations
- Lagrange error bound
- Radius and interval of convergence of power series
- Finding Taylor and Maclaurin series for a function
- Representing functions as power series and manipulating known series

**LEARNING OBJECTIVES**

- 1 Select a convergence test using a systematic decision procedure and verify its conditions.
- 2 Determine the radius and interval of convergence, including endpoint behaviour.
- 3 Construct a Taylor polynomial and bound the truncation error using the appropriate bound.
- 4 Derive a new series by manipulating a known one through substitution, differentiation, or integration.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Selects and correctly applies a convergence test in 9 of 10 series, with conditions stated.
- Determines intervals of convergence including endpoints in 9 of 10 trials.
- Applies error bounds correctly in 4 of 5 tasks.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                                     | RECOMMENDED RESPONSE                                                                                                              |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>The <math>n</math>th term test is used to conclude convergence.</b>            | It can only establish divergence. If the terms tend to zero the test is inconclusive — say so explicitly and choose another test. |
| <b>Endpoints of the interval of convergence are not tested separately.</b>        | The ratio test is inconclusive at the endpoints. Each must be substituted and analyzed independently.                             |
| <b>The alternating series error bound is applied to a non-alternating series.</b> | The conditions must hold. Use the Lagrange bound for the general case.                                                            |

| MISCONCEPTION                                                                               | RECOMMENDED RESPONSE                                                                                                                       |
|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Taylor series are constructed term by term when a known series could be manipulated.</b> | Derive from the five standard series by substitution, differentiation, or integration. This is faster and is what the examination expects. |

**TEACHING NOTE**

*Series accounts for roughly a sixth of the BC examination and is where BC students most often lose points. Give the student an explicit written decision tree for test selection, require the test to be named and its conditions verified before any work, and drill the five standard Maclaurin series to automaticity.*

**ASSESSMENT AND MASTERY POLICY**

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| ASSESSMENT                             | WHEN                                   | WHAT IT MEASURES                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Placement Assessment</b>            | Before enrollment                      | Precalculus fluency including parametric, polar, sequences and series; determines AB versus BC recommendation.                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Unit Check</b>                      | End of each unit · 60 min              | AP-format items in both calculator and non-calculator sections, scored with AP rubrics.                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Justification Portfolio</b>         | Continuous                             | Written justifications scored against AP rubric language.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Series Decision Procedure Check</b> | Weekly during Unit 10                  | Test selection and condition verification without computation; isolates the skill that determines Unit 10 performance.                                                                                                                                                                                                                                                                                                                                                           |
| <b>Full Practice Examination</b>       | Three times: mid-year, February, April | Complete timed BC examinations, scored with AB subscore analysis.                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Examination Format</b>              | AP Exam · May                          | 3 hours 15 minutes. Section I: 45 multiple-choice (Part A, 30 questions, no calculator, 60 min; Part B, 15 questions, graphing calculator, 45 min). Section II: 6 free-response (Part A, 2 questions with calculator, 30 min; Part B, 4 questions without, 60 min). Sections weighted 50/50. An AB subscore is reported alongside the BC score. In 2026 the multiple-choice section is administered digitally in Bluebook and the free-response section is handwritten on paper. |

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

## PACING OPTIONS

Ուսումնական տեմպ

| TRACK                        | CADENCE                         | EXPECTED COMPLETION                                                                                            |
|------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Standard BC Year</b>      | 3 lessons/week, 75 min          | Full course September to April with April dedicated to examination preparation.                                |
| <b>BC after AB</b>           | 2 lessons/week                  | Units 6 (extended techniques), 7 (logistic and Euler), 8 (arc length), 9, and 10 only; approximately 22 weeks. |
| <b>Support / Co-Seated</b>   | 2 lessons/week alongside school | Paced to the student's school AP course, with emphasis on Units 9 and 10.                                      |
| <b>Examination Intensive</b> | 3 lessons/week, 8 weeks         | February to April, weighted toward series and polar.                                                           |
| <b>Summer Head Start</b>     | 4 lessons/week, 8 weeks         | Units 1–4 completed before the school year — strongly recommended for BC.                                      |

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

## HOMEWORK AND INDEPENDENT PRACTICE

Տնային աշխատանք և ինքնուրույն պարապմունք

Ninety minutes per lesson. Every assignment includes an AP-format multiple-choice set with calculator status specified, at least one rubric-scored free-response part, and cumulative maintenance. From Unit 9 onward, every assignment also contains two items from Units 1–8 to prevent decay while the new material is absorbing all attention. In Unit 10, every series problem must begin with the named test and its verified conditions before any computation; unnamed work is marked incomplete regardless of the answer.

## PROGRESS MILESTONES REPORTED TO PARENTS

Ուսումնական նվաճումներ

These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- ◆ All AP Calculus AB milestones achieved at BC pace.
- ◆ Integrates by parts and by linear partial fractions.
- ◆ Evaluates improper integrals with explicit limit notation.
- ◆ Applies Euler's method and analyzes logistic models.
- ◆ Computes arc length for planar and parametric curves.
- ◆ Differentiates parametric and vector-valued functions, including second derivatives.
- ◆ Analyzes planar motion: velocity, speed, acceleration, and total distance.
- ◆ Computes polar areas with correctly determined limits.
- ◆ Selects and justifies convergence tests using a systematic procedure.
- ◆ Determines radius and interval of convergence including endpoint behaviour.

- ◆ Applies the alternating series and Lagrange error bounds.
- ◆ Derives Taylor series by manipulating the standard series.
- ◆ Scored 4 or 5 on a full timed practice examination.
- ◆ Completed AP Calculus BC — examination ready.

## MATERIALS AND RESOURCES

Նյութեր և ռեսուրսներ

- Graphing calculator on the College Board approved list — TI-84 Plus CE recommended.
- College Board Course and Exam Description and released free-response questions from at least ten years.
- Bluebook practice access for digital multiple-choice familiarity.
- Վարժարան series decision-tree reference, constructed by the student rather than issued.
- Վարժարան AP Calculus BC problem bank — approximately 3,200 items tagged by unit and calculator status, with a 600-item series sub-bank.

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